A GROUP ACTION FREE ON RESIDUE FIELDS DEFINES AN ETALE EQUIVALENCE RELATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the map j twice (statement and first line of the proof); both displays are part of the counted spans and are preserved verbatim.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-spaces/retained-preamble.tex, sliced mechanically from the chapter's own preamble.tex.

Allowed standard prerequisites (editorial).

- Definition of an equivalence relation on a category fibred in groupoids. Groupoids, Definition groupoids-definition-equivalence-relation, used at proof lines 2511-2513. Supplies the axioms verified at the end of the proof.
- The presentation theorem for algebraic spaces. Theorem theorem-presentation, used at proof lines 2482-2486. Turns an etale equivalence relation into an algebraic space.

Source reference. Original definition 022P: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2714–2714. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original theorem 02WW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3819–3819. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let $U \to S$ be a morphism of Sch_{fppf}. Let G be an abstract group. Let $G \to Aut_S(U)$ be a group homomorphism. Assume*

(*) *if $u \in U$ is a point, and $g(u) = u$ for some non-identity element $g \in G$, then g induces a nontrivial automorphism of $\kappa(u)$.*

Then

$$j : R = \coprod\nolimits_{g \in G} U \longrightarrow U \times_S U, \quad (g, x) \longmapsto (g(x), x)$$

is an étale equivalence relation and hence

$$F = U/R$$

is an algebraic space by Theorem 02WW.

Date: stacks-project:spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. In the statement of the lemma the symbol $\mathrm{Aut}_S(U)$ denotes the group of automorphisms of U over S. Assume $(*)$ holds. Let us show that

$$j : R = \coprod\nolimits_{g \in G} U \longrightarrow U \times_S U, \quad (g, x) \longmapsto (g(x), x)$$

is a monomorphism. This signifies that if T is a nonempty scheme, and $h : T \to U$ is a T-valued point such that $g \circ h = g' \circ h$ then $g = g'$. Suppose $T \neq \emptyset$, $h : T \to U$ and $g \circ h = g' \circ h$. Let $t \in T$. Consider the composition $\mathrm{Spec}(\kappa(t)) \to \mathrm{Spec}(\kappa(h(t))) \to U$. Then we conclude that $g^{-1} \circ g'$ fixes $u = h(t)$ and acts as the identity on its residue field. Hence $g = g'$ by $(*)$.

Thus if $(*)$ holds we see that j is a relation (see Groupoids, Definition 022P). Moreover, it is an equivalence relation since on T-valued points for a connected scheme T we see that $R(T) = G \times U(T) \to U(T) \times U(T)$ (recall that we always work over S). Moreover, the morphisms $s, t : R \to U$ are étale since R is a disjoint product of copies of U. This proves that $j : R \to U \times_S U$ is an étale equivalence relation. $\square$